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Influence of Laser Surface Texturing on Tribological Performance of Tool Steels

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This paper deals with the influence of laser surface texturing on tribological performance of two different tool steels: (a) Böhler K100 produced by conventional production technology and (b) Böhler K390 Microclean® produced by powder metallurgy. The surface of heat-treated tool steels was modified using three different texture patterns. Impact of laser surface texturing was analyzed after different treatments. Furthermore, selected samples were borided to evaluate the combined influence of laser surface texturing and boriding on tribological performance of surface-treated samples. The results show that boriding was not beneficial for wavy surface type, and the steel production technology had a certain influence on the spinning tool. The laser surface texturing had an effect on decreasing the coefficient of friction on the K100 tool steel; however, it did not influence the tribological performance of the K390 Microclean® tool steel. Boriding process was efficient in decreasing the coefficient of friction for both steels. The low coefficient of friction values did not have a positive influence on the wear depth.

Keywords boriding, laser texturing, texture pattern, tool steel, tribology

1. Introduction

Laser texturing is a method which uses an intentionally formed specific surface pattern to modify functional properties of a tool. Laser can create different textures from nano-dimensions to few millimeters. Laser surface texturing can be applied regardless of the type of material of the textured part. The pulse laser texturing influences the surface quality and improves specific tribological applications and grease retention on cold-surface applications (Ref 1-3). The laser creates a desired regular pattern consisting of repeating elements with the aim of creating specific surface properties, where each element is of the size of a few micrometers.

The determination of tribological properties is typically based on various methods. The appropriate method and parameters are chosen close to real conditions in which the functional surface of the component is applied (Ref 4-9).

Metal spinning technology requires abrasion resistant and tough tools to achieve a sufficient steel sheet deformation during forming. The highest loading force is applied at the tool tip and as such is needed to improve the tool service lifetime. Some works (Ref 10-14) present the characterization and practical usability of metal spinning process and others (Ref 15-18) present the process design influence on treated material. Tool steels are produced by various technologies. Variation in properties of steel with identical nominal chemical composition is primarily due to different carbide size and their distribution in the microstructure (Ref 19-22). It is therefore important to mutually compare the tribological performance of tool steels with coarse and fine carbides after laser texturing and surface treatment applied.

Most degradation mechanisms, such as wear, fatigue and corrosion, start at the surface (Ref 23, 24). Improvement of surface properties of parts is currently possible via deposition of thin, hard and wear-resistant coatings. At present, different methods are used to improve functional surfaces, such as laser hardening, laser cladding, boriding, plasma enhanced nitriding, coating process (PVD, CVD, DLC), laser texturing, etc. (Ref 25-35).

In the present paper, two different tool steels were investigated: (a) Böhler K100 produced by conventional metallurgy process and (b) Böhler K390 Microclean® produced by powder metallurgy. Furthermore, selected samples have been borided to evaluate the combined influence of laser surface texturing and boriding on tribological performance.

2. Experimental Part

Two different tool steels, K100 and K390 Microclean® produced by Voestalpine Böhler Edelstahl GmbH & Co KG were used in the experiment. Nominal chemical composition of analyzed tool steels is shown in Table 1. Both tool steels were primarily defined as cold working tool steels. After appropriate heat treatment, these steels possessed a high hardness, abrasion resistance, appropriate toughness, good resistance to high pressure loads and deformations, as well as good dimensional stability and proper machinability in annealed state (Ref 36, 37). The K100 tool steel was produced by conventional metallurgy methods, while K390 Microclean® was produced by powder metallurgy.

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Table 1 Nominal chemical composition of analyzed tool steels (Ref 33, 34)

Chemical composition, wt.%	C	Si	Mn	Cr	Mo	V	W	Co
K100	2.00	0.25	0.35	11.50	…	…	…	…
K390 Microclean®	2.47	0.55	0.40	4.20	3.80	9.00	1.00	2.00

Table 2 Technological parameters of laser texturing device used for surface modification of analyzed tool steels

Material	Pulse separation, µm	Frequency, kHz	Movement speed, mm s^{-1}	Power, W
K100	10	70	1500	48.7
K390 Microclean®	10	70	1500	51.5

Rods with 16 mm in diameter and 150 mm in length were used. In order to obtain the appropriate mechanical properties of analyzed tool steels, hardening and tempering were applied. Hardening temperature was 950 °C for K100 tool steel, and 1050 °C for K390 Microclean®, respectively. The tempering followed the hardening process immediately, and K100 tool steel was tempered at 250 °C, while K390 Microclean® tool steel was tempered at 550 °C. After the heat treatment, the hardness was measured by Rockwell method. The hardness achieved was 59.3 ± 0.6 HRC for K100 tool steel, and 57.3 ± 1.2 HRC for K390 Microclean® tool steel, respectively. Heat-treated rods were cut into small disks by using a precision cutting saw. The experimental surface was grinded and polished to an average roughness $R_a \approx 0.018$ µm as determined by Zeiss Surfcom 5000. Finally, the laser surface texturing was realized using the 5-axes SAUER Lasertec 80 Shape. The technological parameters of laser texturing are shown in Table 2.

Selected samples were powder-borided using Durborid®. The K100 tool steel was borided at 850 °C and the K390 Microclean® at 950 °C, respectively. The holding time was 1 h at the peak temperature for both tool steels. The expected thickness of the boride layer was up to 20 µm, as was experimentally determined during preliminary experiments.

The ball on disk method was used for tribological testing of experimental samples. The experiments were performed at the Slovak Academy of Sciences, the Institute of Materials Research in Košice using Bruker UMT 3 universal tribometer. The aim was to determine the friction coefficient between the functional surface and the friction ball. The friction ball was produced from 100Cr6 hardened steel and had 6 mm in diameter. Parameters of ball on disk test are shown in Table 3. Parameters of tribology test were determined from simulation of forces acting during the metal spinning process. During ball on disk test, both the tangential component of friction force (F_x) and normal component of friction force (F_z) were collected. The friction coefficient was calculated as a relative ratio of F_x and F_z and recorded every 0.01 s. Wear depths were measured by two physically different methods: (a) the laser confocal microscopy (Zeiss LSM 700) and (b) the Zeiss Surfcom 5000 equipped with stylus with 2 µm tip radius.

The weight loss after tribological tests was determined by the precise laboratory balance Mettler Toledo with accuracy ± 0.0001 g.

3. Results

3.1 Microstructure Analysis

Figure 1 presents cross-sectional micrographs of both tool steels. The carbide size and distribution were significantly different (Fig. 1). Large carbide particles were observed in K100 tool steel produced by conventional technology. Contrarily, the matrix of K390 Microclean® contained a homogeneous distribution of small carbide particles, produced during the powder metallurgy production process.

3.2 Laser Surface Texturing

Three different types of surface texture patterns (Fig. 2) were prepared by hatching method. The modeled and manufactured texture patterns are compared in Fig. 2.

The T1 texture consisted of dents with a diameter of 0.1 mm and a depth of 0.05 mm spaced 0.5 mm apart. The density of the textured surface was 6.3% (Fig. 2a). Dents are meant to retain grease and worn material. It is expected that this should reduce friction, and a tribological pair of lubricant surface should be created which will move over the textured surface.

Texture T2 was designed according to texture 1 but in reverse (Fig. 2b). The surface fill density reached 93.7%. The texture was composed of hemispherical projections, similar to texture 1. The diameter of the hemispheres was 0.1 mm, the height was 0.05 mm and they were positioned 0.13 mm apart. The T2 texture provides a good wettability of the surface, and the surface of contact is only on the top of the protrusions.

Texture T3 was an experimental texture. It was built on the assumption that the lubricant would be trapped inside the ring. The texture covered approximately 38% of the surface (Fig. 2c). The texture consists of the protruding intersections separated by 0.1 mm. The total distance of the median centers was 1 mm. The depth of the internal grooves was 0.1 mm.

3.3 Tribological Performance of Textured Surfaces

Friction coefficients were measured as a function of time. The results are presented in Fig. 3 for all textured surfaces. The friction coefficients varied with tested material. For non-borided experimental materials, T1-textured pattern had a higher friction coefficient, while the remaining two textured patterns and non-textured surfaces were comparable. In the case

of borided steels, the results were different (Fig. 4). All three textured patterns were comparable for K100 tool steel. Sample with the non-textured surface manifested a gradual increase in friction coefficient value during the friction test. Contrarily, K390 Microclean® tool steel exhibited a smaller friction coefficient for non-textured surface.

The best friction coefficient evolution during friction test was found for borided K100 tool steel with T1 texture. Texture T2 had slightly higher values of friction coefficients for borided surfaces of K390 Microclean® steel than other samples, which were rather similar. The comparison of T3 texture showed again higher values for borided K390 Microclean®. Other samples with T3 texture pattern were mutually comparable. Boriding did not have a positive influence on the service duration of tested surfaces as the friction coefficient was increasing with loading time. Contrarily, untreated surfaces exhibited better values of friction coefficient for both materials analyzed.

Figure 5 shows the average friction coefficients of tested texture patterns as a function of tool steel type and surface treatment. The results show that K100 had the lowest average friction coefficient for all texture types after boriding. In the case of non-borided samples, the textures T3 and T2 were found to be similar. The K390 Microclean® tool steel had the lowest average friction coefficient for T2 and T3 textures and non-textured samples. All borided surfaces had slightly higher values of friction coefficients. A higher average friction coefficient was also determined for non-borided sample with T1 texture.

Next, the Hertz pressure was calculated by ball on flat surface method. Based on material constants of materials in mutual contact and normal force applied, the maximal Hertz contact pressure $p_{max.}$ = 2623.5 MPa was calculated, which caused the maximal shear stress $\tau_{max.}$ = 816.4 MPa evolved in 0.054 mm of depth. Diameter of circular contact area was calculated to be 0.226 mm. The depth resolution of stress from the contact surface is shown in Fig. 6(a) (Ref 38).

Hertz pressure for ball on ball method was calculated for comparison. If the contact surface is very small (laser textured surface T2, T3), then the contact can be described as a contact area between the sphere with 6 mm diameter and sphere with diameter of 0.1 mm (tip of textured patterns). In this case, the maximum Hertz contact pressure was 40,653.7 MPa which caused maximum shear stress 12,650.3 MPa acting in 0.014 mm depth. The circular contact area diameter was calculated to be 0.057 mm. The distribution of the stress is shown in Fig. 6(b) (Ref 38).

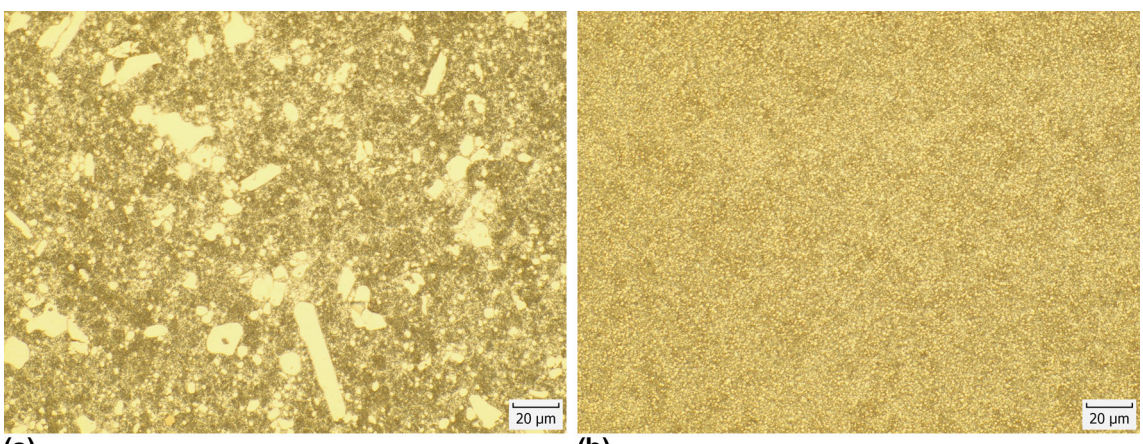
The Hertz contact pressure as well as the shear stress after laser texturing were significantly higher than the ultimate strength of the material. As a consequence, the texture tips were in the first stage of tribological test intensively loaded and deformed, which led to both higher wear depths and higher weight losses.

Table 3 Parameters of tribological test

Method	Ball on disk
Applied load, N	70
Sliding velocity, m s^{-1}	0.1
Frequency of data write, s	0.01
Track length, m	200
Track radius, mm	3.5
Ball diameter, mm	6
Time of load, s	2000

3.4 Laser Confocal Microscopy and Profilometry of Worn Surfaces

Topography of samples with wear tracks, obtained by laser confocal microscopy, and depth profiles, obtained by Surfcom 5000, are shown in Fig. 7 for the K390 Microclean® tool steel. Similar results were obtained for the K100 tool steel. The measured wear depths are compared in Fig. 8. A very low depths were found on borided K100 tool steel with T1 texture and borided non-textured surfaces: (3.85 ± 0.01 μm) and (2.81 ± 0.01 μm), respectively. Sample with T1 texture without boride layer had a low wear depth too (8.95 ± 0.50 μm); however, it was at least two times higher than for the borided specimen. The lowest wear depths in the case of K390 Microclean® were found for borided T1-textured pattern (0.67 ± 0.01 μm), borided non-textured surface (1.10 ± 0.02 μm), non-borided and non-textured (0.81 ± 0.01 μm) surface, respectively. The remaining samples had very high wear depths.



(a) (b)

Fig. 1 Microstructure of analyzed tool steels. (a) K100. (b) K390 Microclean®

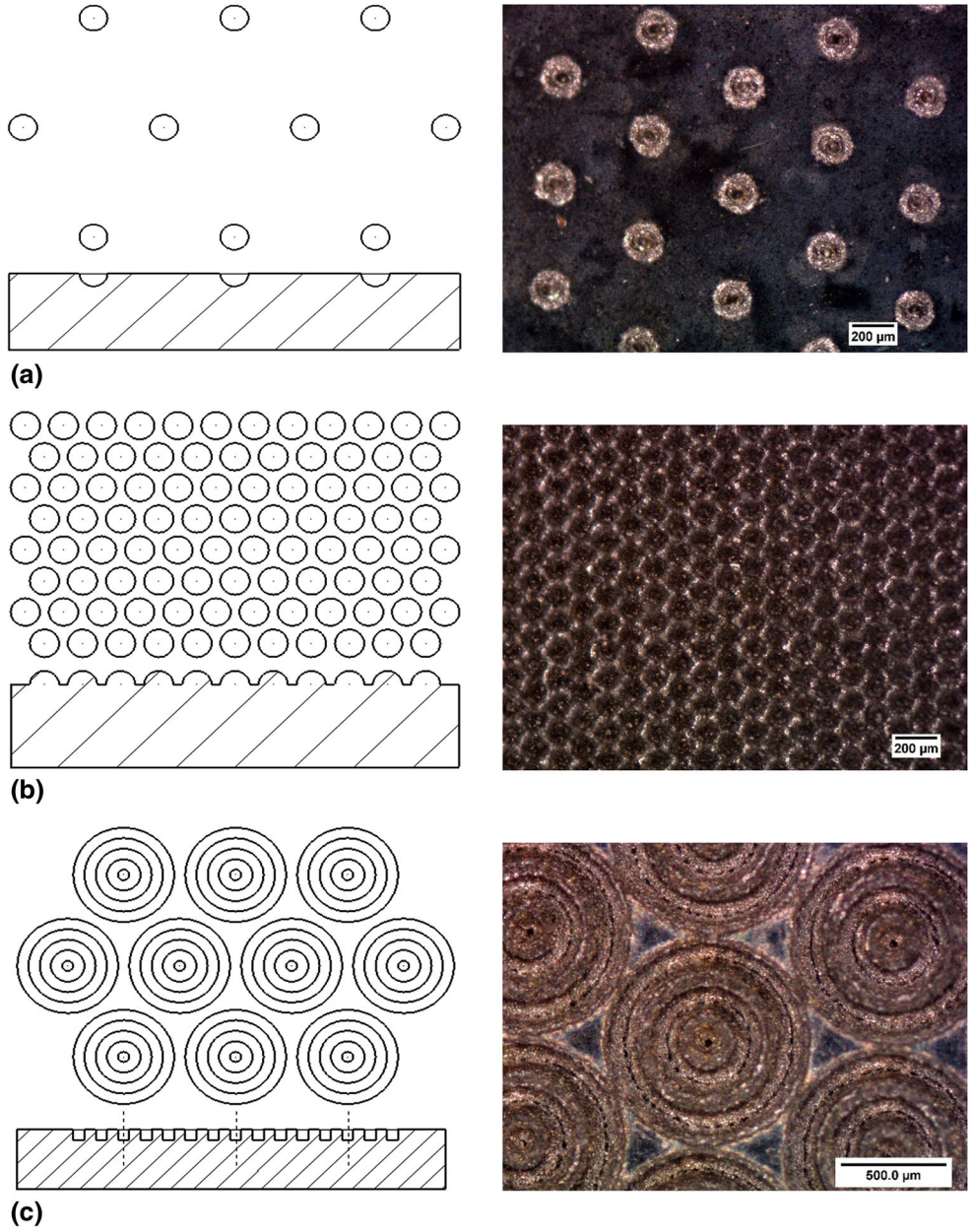


Fig. 2 Laser-textured patterns. The left-hand side image represents the model scheme of the texture, while the right image documents the real textured pattern after laser treatment of the polished tool steel surface. (a) textured pattern T1. (b) Textured pattern T2. (c) Textured pattern T3

3.5 Weight Loss Analysis

The measurement of the weight loss after tribology test yielded the average value for each analyzed texture pattern, which is documented in Fig. 9. All non-textured surfaces had very low weight losses values, and their values were comparable. The highest weight loss was found for textures T2 and T3 on borided 390 Microclean® tool steel. This can be explained by the fact that these two texture patterns were prone to high wear localized at the surface tips of textures during exposure due to their hardness after surface laser treatment and high Hertz pressure. Texture T2 negatively influences the tribology of K100 tool steel after boriding too. Other samples had comparable results.

3.6 Discussion

Results reported above show that the applied textures had a remarkable effect on tribological properties. Furthermore, the evaluated parameters demonstrate that the performance of textures was affected by the tool steel substrate. As can be seen in Fig. 3, the coefficient of friction (COF) determined for texture T1 increased over time for both tool steels while the laser-treated area was only 6.3% out of total sample area. Increase in COF has not led to higher wear (Fig. 9). The wear depth was also relatively small (Fig. 9) for both steels. This is likely caused by a synergic effect of deformation in contact spots and total area surrounding pits hardened by laser remelting which are both directly determined by texture

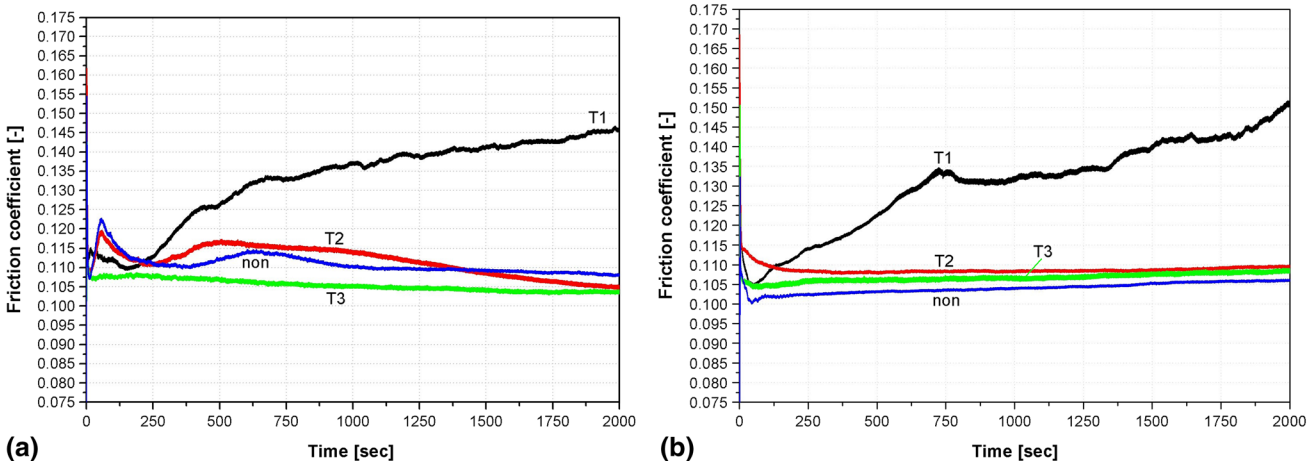


Fig. 3 Friction coefficient of textured samples as function of time. (a) K100. (b) K390 Microclean®

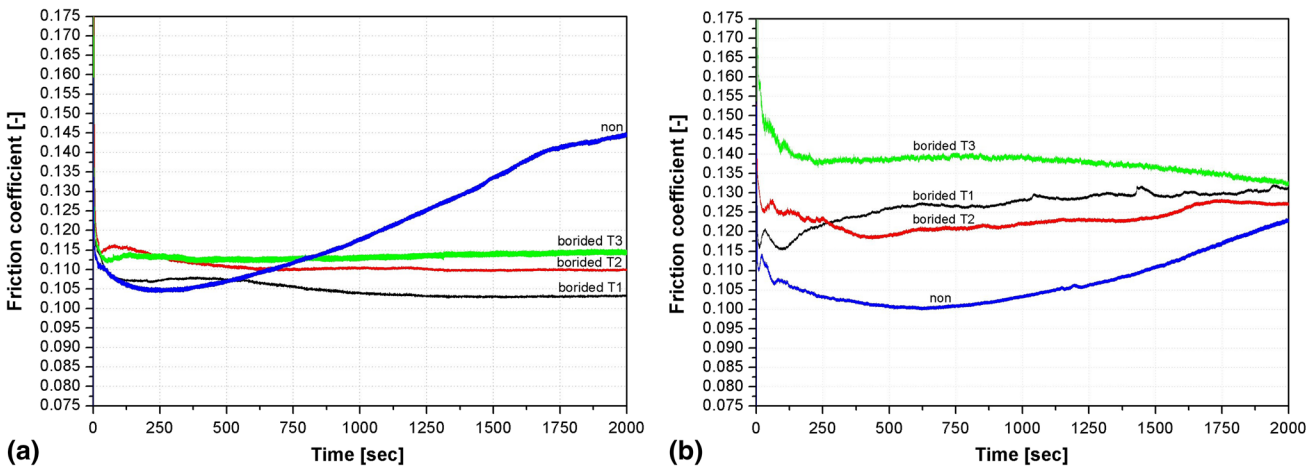


Fig. 4 Friction coefficient of textured and borided samples as function of time. (a) K100. (b) K390 Microclean®

pattern. This is in contrast to the non-textured surfaces where only a plastic deformation prevailed (Fig. 7a and b). The influence of T1 texture is also apparent in Fig. 5 for both tool steels.

Texture *T*2 represents approximately 93.7% of surface area influenced by laser beam. The COF values were found to be relatively low and comparable for both tool steels (Fig. 5). Only a small increase in weight loss (Fig. 9) was observed for non-borided *T*2 textured surfaces when compared to *T*1 counterparts. Surfaces with *T*2 textures exhibited the highest wear depth for both tool steel substrates and treatments (Fig. 8). As a consequence in T2 texture pattern high local loads were imposed on protuberances (Hertz pressure $\sim$ 40,653 MPa) leading to their damage and fragmentation. It is assumed from Fig. 7(c) that a significant portion of fragments filled the interspaces of protuberances while the remaining debris were removed from the surface by centrifugal forces acting during the test. This is the reason for the relatively low COF value while the wear was high.

The texture pattern *T*3 was devised as an intermediate step between *T*1 and *T*2 textures because of its 38% of surface area affected by laser texturing. The weight loss was the highest in this case for both tool steel substrates (Fig. 9). The wear depth was found to be lower than in the case of *T*2 texture but higher than for the *T*1 textured and non-textured surfaces (Fig. 8). The highest COF value was determined for borided *T*3 textured surface of K390 tool steel (Fig. 4b). The wear depth was, at the same time, relatively high, as documented in Fig. 8.

The lowest wear depths (Fig. 8) and weight losses (Fig. 9) were found for non-textured surfaces. However, non-borided and non-textured K100 tool steel exhibited highest COF values overall (Fig. 5a).

Based on the above-presented results it can be stated that the surface area affected by laser treatment induced remarkable variations in the tool steel tribological behavior. Low COF values were neither sufficient to yield low weight losses nor to yield sufficiently low wear depths. The overall performance of the sample depends on a complex interplay of several variables, thereby determining the tribological behavior of the material with given chemical composition and heat treatment.

4. Conclusions

The combined influence of laser surface texturing and boriding on tribological properties has been analyzed in the present work. The results achieved during tribological testing of non-textured surface of two different tool steels prepared by various production technologies were compared. The experi-

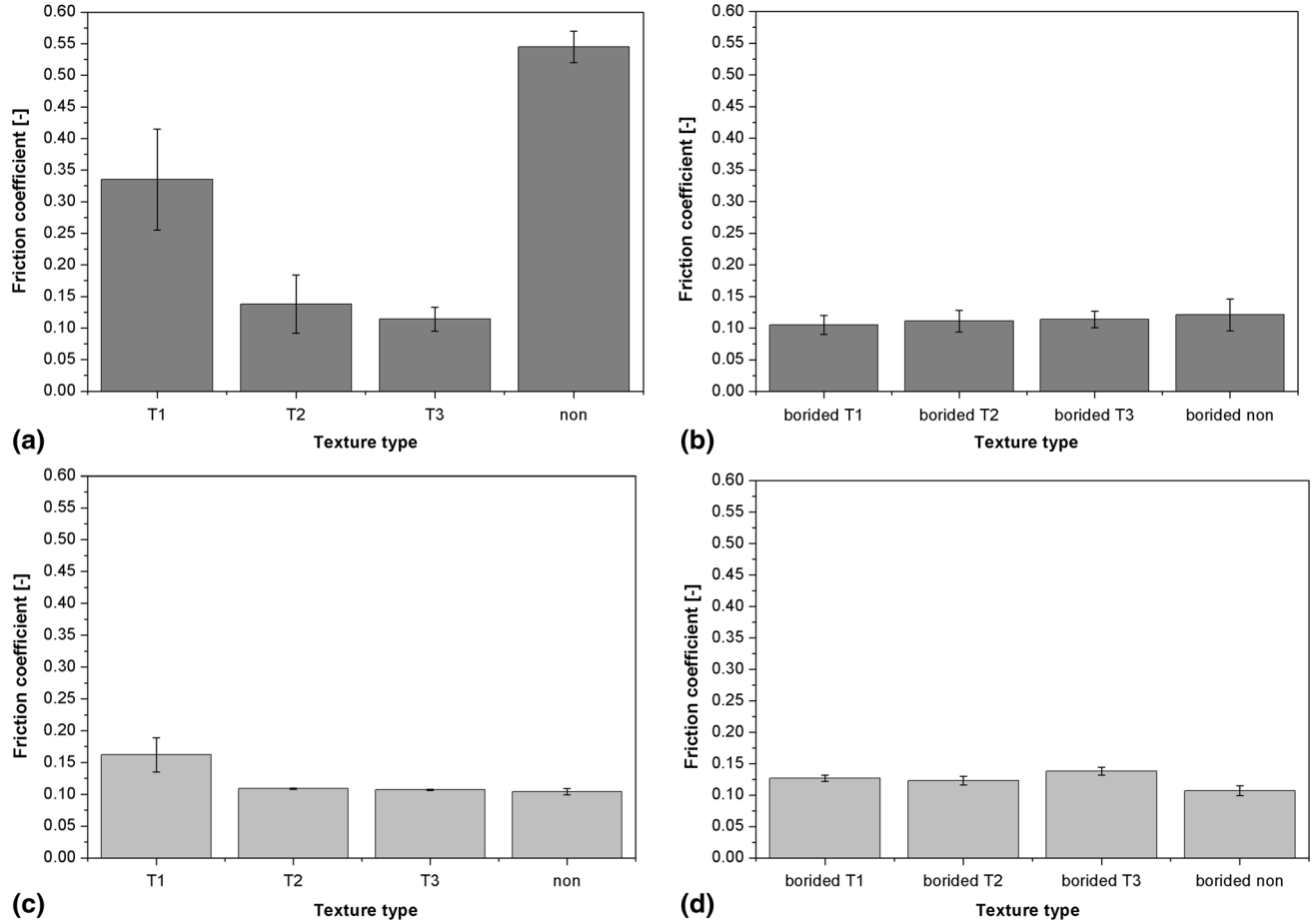


Fig. 5 Friction coefficient for analyzed samples. (a) K100 textured. (b) K100 textured and borided. (c) K390 textured. (d) K390 textured and borided

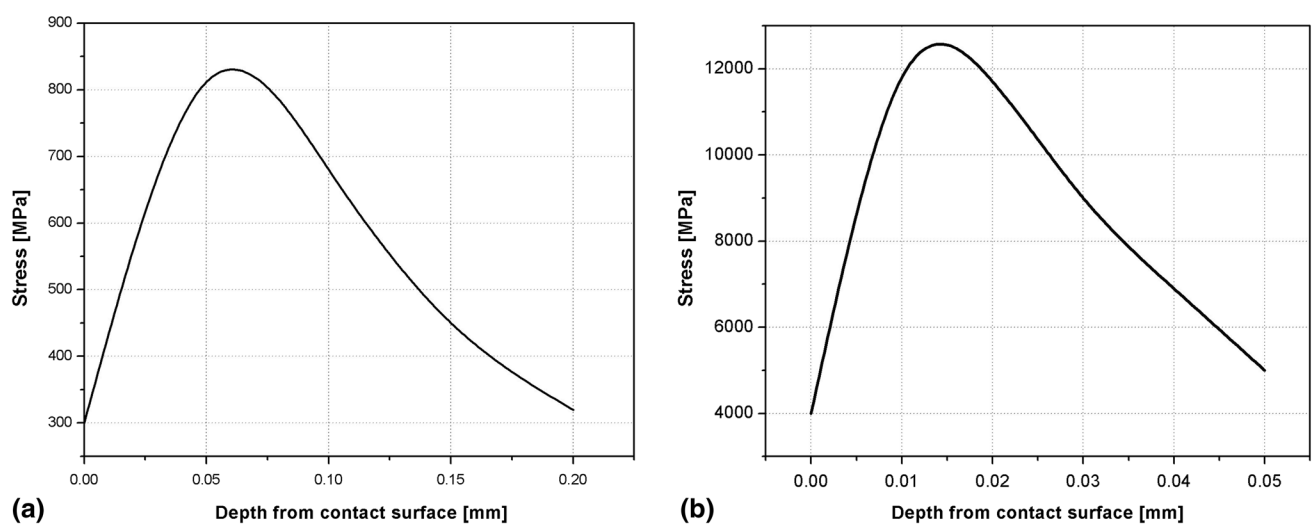


Fig. 6 Stress change with depth from contact surface (Ref 38). (a) sphere on flat plate. (b) Ball on ball

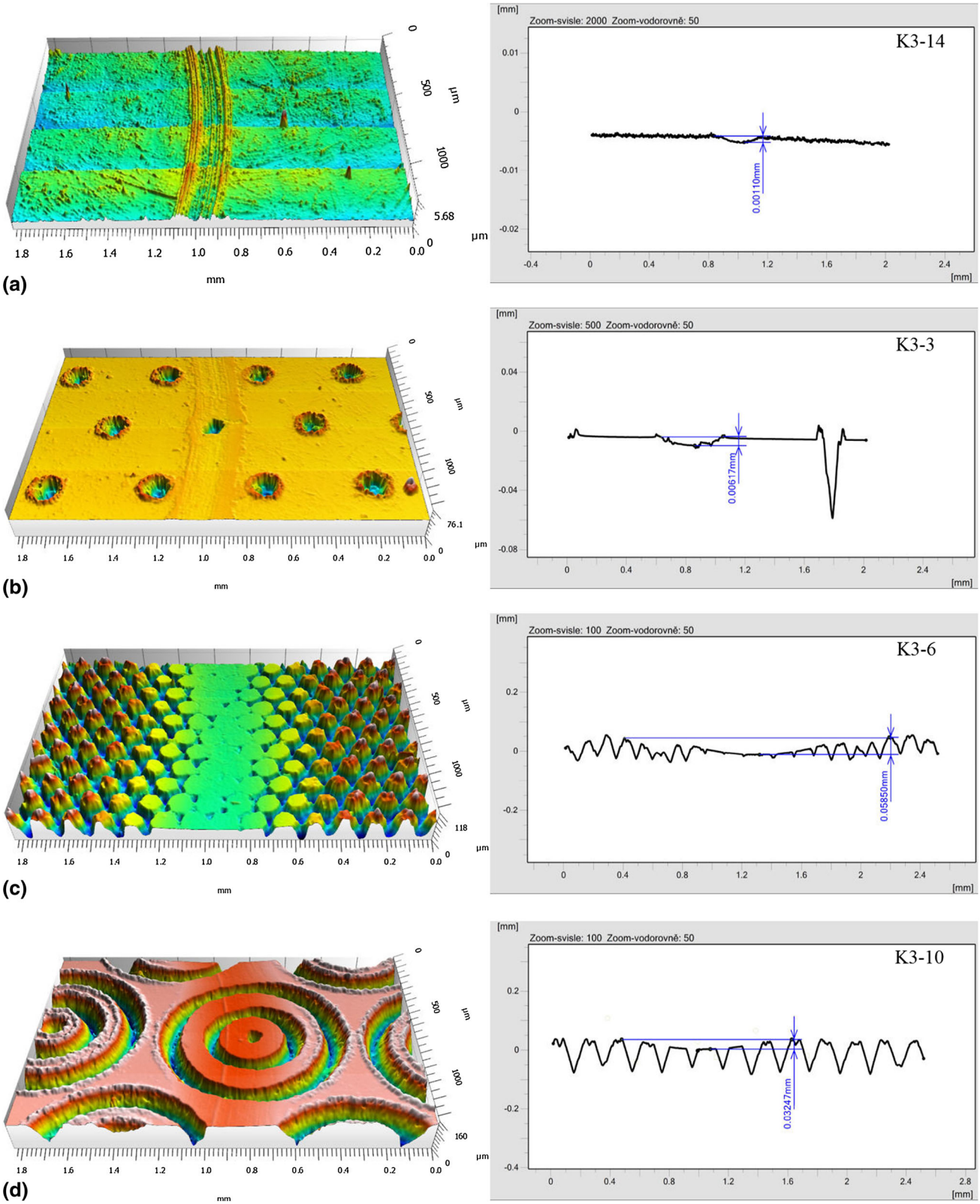


Fig. 7 Deformed sample surfaces after tribology test: left: overview; right: depth profile. (a) Non-textured K390 Microclean®. (b) T1 texture on K390 Microclean®. (c) T2 texture K390 Microclean®. (d) T3 texture K390 Microclean®

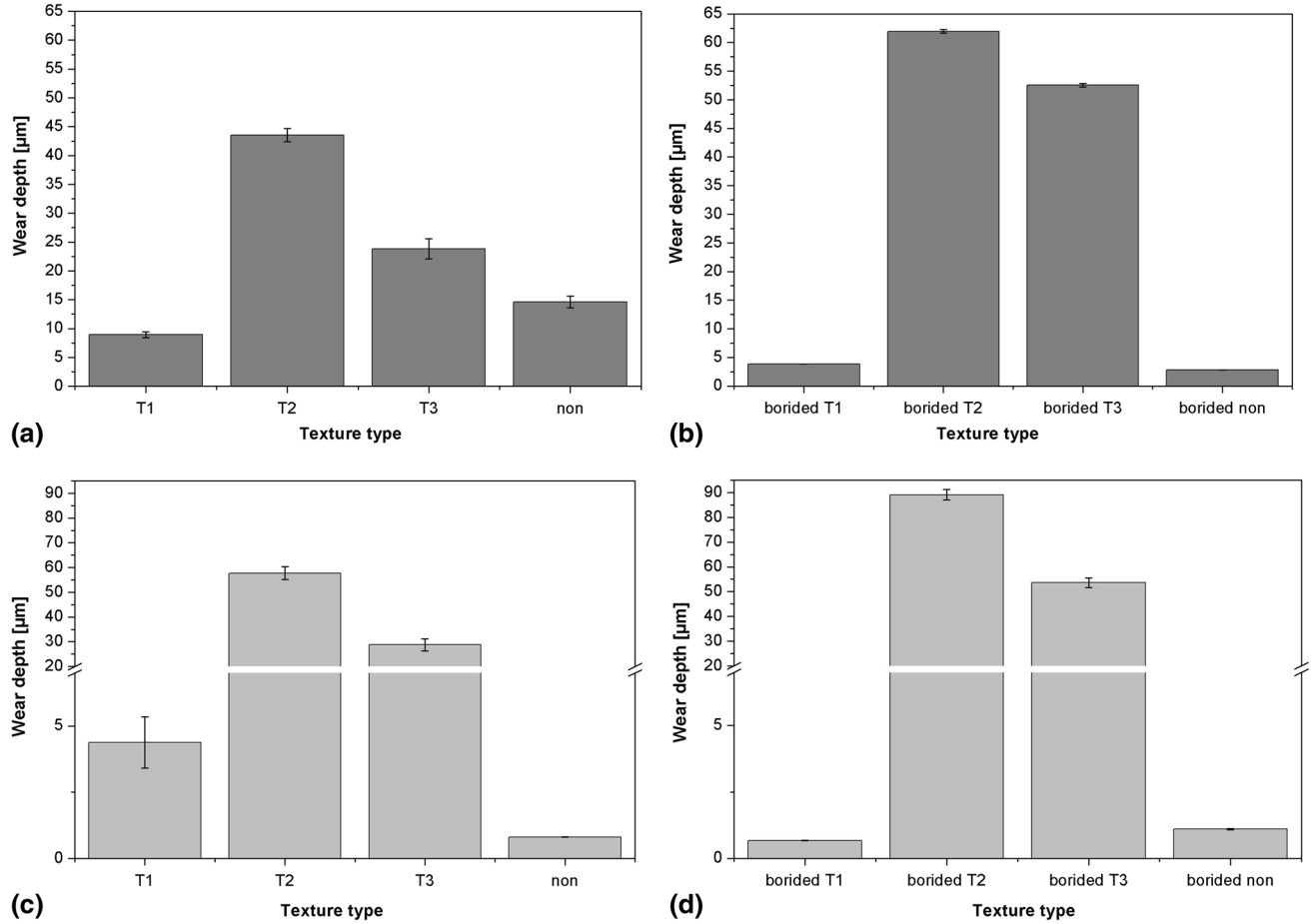


Fig. 8 Wear depths of analyzed samples. (a) K100 textured. (b) K100 textured and borided. (c) K390 textured. (d) K390 textured and borided

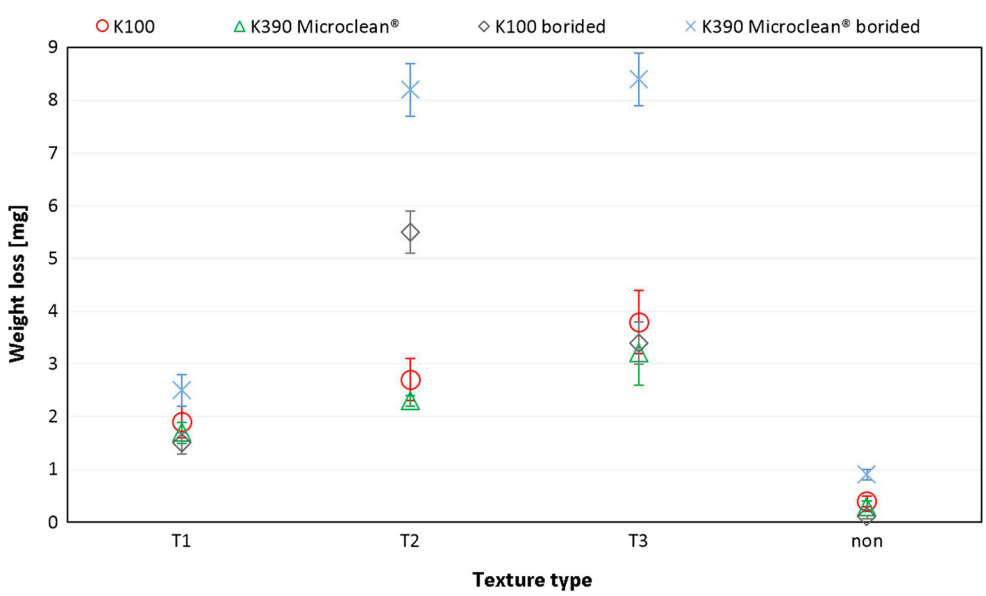


Fig. 9 Weight loss dependence on texture type

mental results obtained allowed us to draw the following conclusions:

1. Laser surface texturing applied on a larger fraction area of K100 tool steel surface had a positive effect on decreasing the coefficient of friction. However, higher weight losses of material as well as higher wear depths were observed in comparison with non-textured surfaces and surfaces with smaller fraction area of textured surfaces.

2. Laser surface texturing had no significant effect on K390 Microclean® tool steel. COF remained nearly unaffected. Weight losses and wear depths were higher for surfaces with higher fraction areas of texturing.
3. In general, boriding had a positive effect on COF (Ref 39-41). It was, however, found to be more suitable for smooth continuous surfaces or surfaces with smaller fraction of texturing.
4. Low COF values were not sufficient to yield small weight losses and small wear depths. Coefficient of friction may not be decisive in the case of textured surfaces because of higher amount of wear.

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